



Bluebike Summer Usage Pattern: Determinants of a Station's Popularity

Ashley Kang Wellesley College Class of 2025 Data Science Major Capstone

Introduction

Background:

Bluebike is a public transportation system centered in the Metro Boston area that offers users over 4,500 bikes from more than 480 stations, ranging from Arlington to Watertown.

Research Question:

What station characteristics and demographic factors contribute to the popularity of Bluebike stations in Massachusetts during the summer months?

Data Cleaning

Sources:

- Bluebike Station Data
 - Indiv. station information: total docks, municipality, seasonal status, location (lat. / long.)
- Bluebike 2020-2024 trip history for the summer months (June through Sept.)
 - Indiv. trip information: total duration, user's membership status, start/end station
- Higher Educational Institution Data
 - US institution information: name, location (lat. / long.), state abbreviations
- MBTA Train Station Data
 - Individual station information: name, location (lat. / long.)
- Massachusetts Weather Data
 - avg. minimum temperature, avg. maximum temperature, avg. precipitation
 - Used separate datasets (above/below the Charles) due to limited datasets
- New variables:
 - Trip Duration (Numeric)
 - Trip Displacement (Numeric)
 - Population (Numeric)
 - n_institution_1km (Numeric): institutions within 1km of the station
 - n_mbtta_1km (Numeric): MBTA stations within 1km of the station

Data Manipulation:

- Dimensions: 8800 observations x (27 predictors + 1 response)
- Aggregated monthly
- Removed rows with missing and non-imputable station information
- Tested Full, AIC, and BIC first-order models for model selection

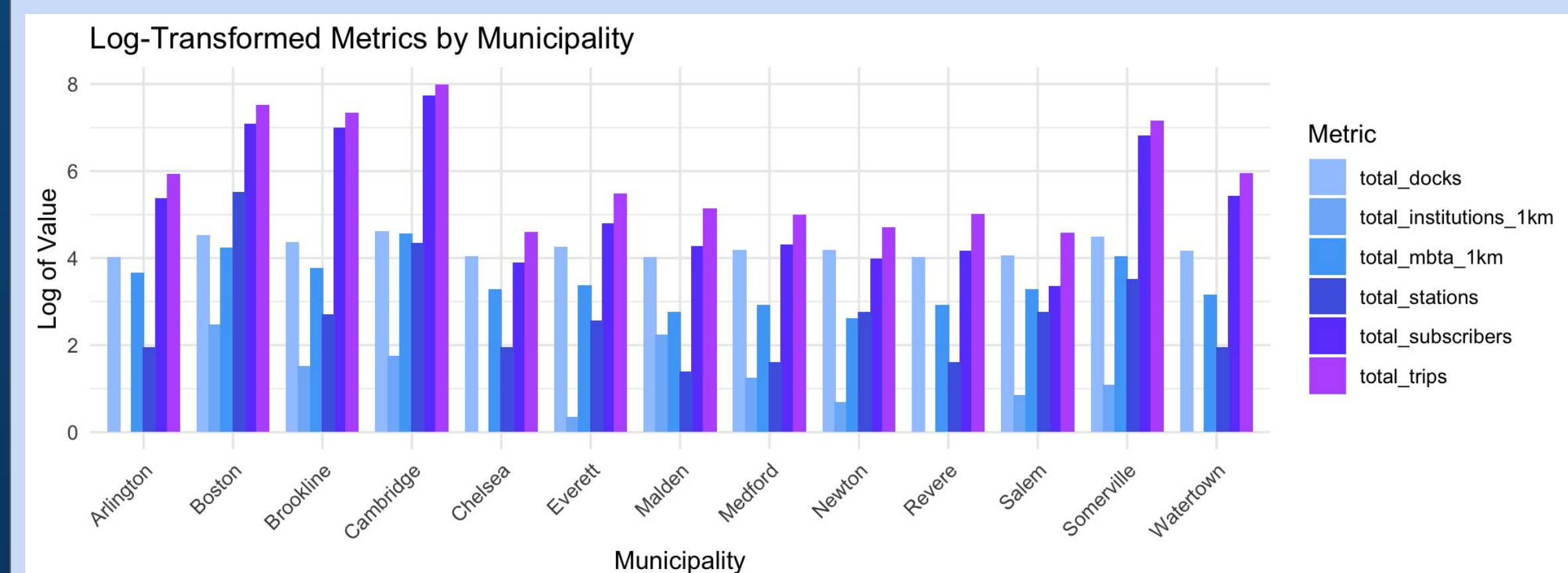


Figure 1: Barplot by municipality showing the distribution of five key factors logged along with total trips as the response variable. We can see that most of the factors are positively correlated with the response variable total_trips.

Methodology

STEP 1: Check multicollinearity: remove the 'population' variable due to the VIF score being 53

STEP 2: Identify the best first order model by comparing multiple metrics, including the 10-fold CV

	Full Linear	AIC	BIC	BIC no outliers	AIC interaction	AIC interaction no outliers	BIC interaction
RMSE	18.798	18.829	18.831	13.922	15.071	11.155	15.470
R^2	0.965	0.965	0.964	0.973	0.977	0.984	0.976
MAE	11.539	11.544	11.515	9.326	9.474	7.621	9.742
p	26	23	21	21	146	146	65

Table 1: Table for comparing the six models. We can see that the all three metrics don't vary much by the non-interactive model, but the AIC model with interaction tends to be slightly better with the lowest RMSE, highest R squared, and lowest MAE. The threshold for determining the outliers is 4/n.

STEP 3: Check the assumptions for the best model, which is the AIC interaction model with outliers removed

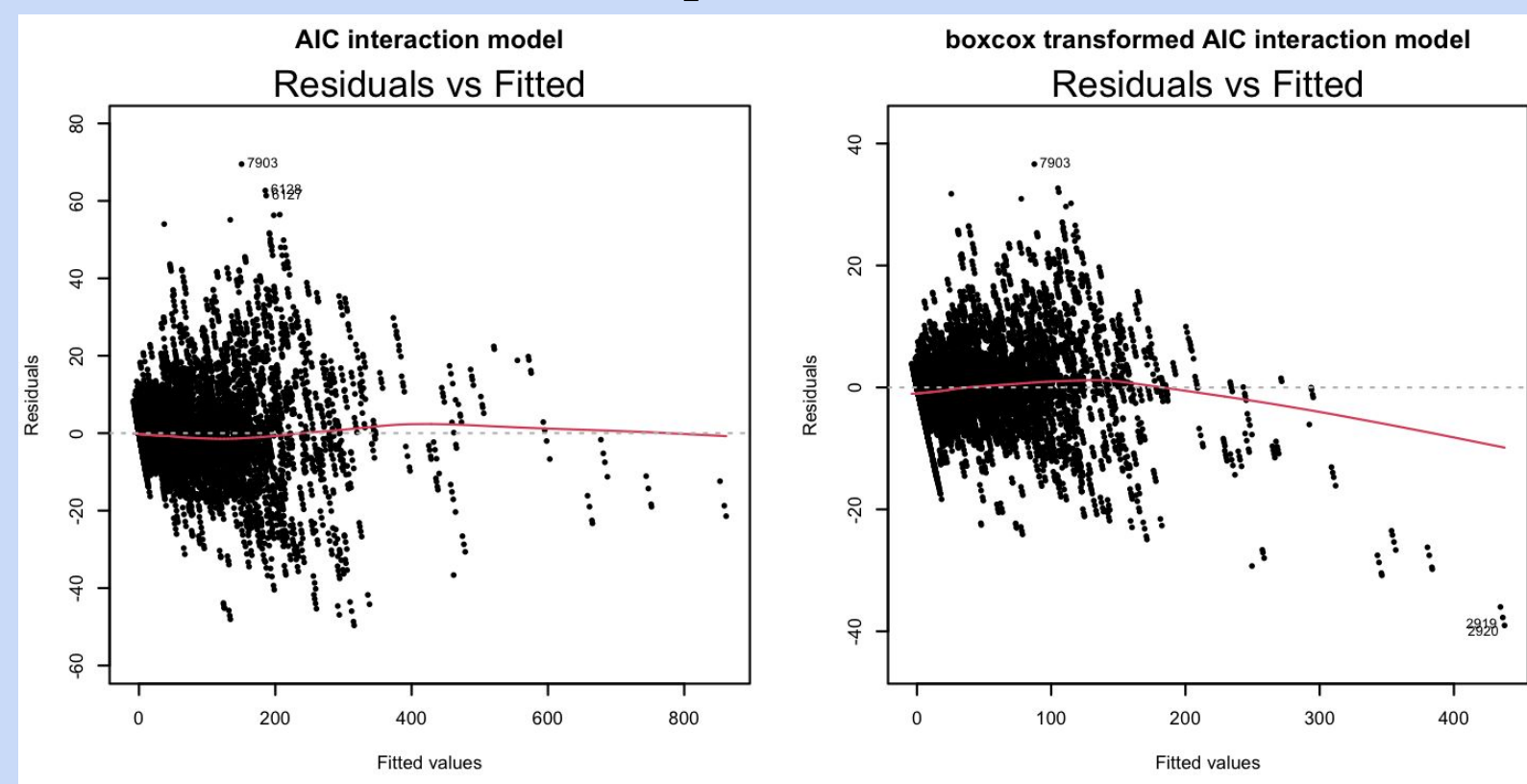


Figure 2: This is a plot of the AIC interaction model with outliers removed and transformed using the Box-Cox method. This transformation is better as the vertical spread is more stable compared to the one before transformation. The larger cluster observed in the plot corresponds to a higher concentration of observations in Boston, Cambridge, and Somerville. This grouping is expected, given the greater number of data points from these municipalities. Despite this, the residuals appear to be evenly distributed around the zero line across the range of fitted values, suggesting that the model satisfies the constant variance assumption.

STEP 4: Conduct a partial F test to compare the AIC interaction model with the first-order BIC model (these two models are nested) to determine if the interaction terms are necessary.

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
BIC	8858	3147771				
AIC interaction	8733	1982167	125	1165605	41.083	< 2.2e-16

Table 2: This is the table of the anova result of the first order BIC model and the AIC interaction model. From this table, we can calculate the partial F-stat, which is 41.0832, which ends up having the p value of 0.

Results and Model

Based on the observations, the interactive model appears to be the best model for predicting the popularity of a station.

Predictor	Coefficient	Predictor	Coefficient	Predictor	Coefficient
Intercept	5628000	n_institutions_1km:n_mbtta_1km	0.1646	n_institutions_1km:Long	-54.46
Longitude	79180	n_institutions_1km	-4055	avg_TMIN:avg_TMAX	-0.09651
Lat:membership_total	2.868	Latitude	-109600	Lat:Long	-1537
Long:membership_total	2.801	n_mbtta_1km:membership_total	-0.00719	total_docks:n_mbtta_1km	-0.03006

Table 3: Table of my final model's predictors and coefficients (only included the predictors with the most significant p-value) Positive coefficients are highlighted in blue, and negative coefficients are highlighted.

Discussion and Conclusion

- Discussion:** The weather related variables do not have an impact on the model. This could be due to the consistent summer temperatures over the years, as the temperature during the summer does not vary significantly over years.
- Limitation & Future Work:** The relationship between the predictors and the outcome may be non-linear, which standard linear models may not fully capture. I will propose the Random Forests method to better capture complex, non-linear patterns in my predictors. Also for a better prediction, I can add tourist attractions as a factor since people tend to visit Boston to tour the area.